Final Project

The final project for this class is completely open ended. You can pick just about any topic you wish so long as there is some connection to statistical machine learning and its algorithms.

- **Group Project.** You can work individually or in small groups of two or three, although larger groups need to be justified by larger projects.

- **Deadlines.** Please email me and Haoyun (in plain text) a paragraph or two outlining your project as soon as you know what you want to do, but no later than Friday, November 18, 2022. **The submission deadline is 11:59 pm of December 14, 2022.** Note that the grades due time for Fall 2022 is December 20. I strongly advise starting early on your project. Running experiments takes time, as does thinking about research problems. It is very important that you get everything to me on time so that I can turn in final grades on time.

- **Suggestions.** For your project, you should start by doing some reading on a topic, and then you might run an experiment, or try to simplify or improve or extend the result, or you might try applying an algorithm to a particular application, or you might think about how two different approaches or algorithms are related to each other. Or you can do something different from any of these. Examples of possible types of projects:

  - Build or apply a machine learning program tailored to a particular application coming from some other field such as natural language processing, engineering, bioinformatics or information retrieval.

  - Think about what the consequences are of changing one of the standard learning models. Does learning become easier or harder? What are the

general properties of the new model? Can you think of new algorithms for interesting problems in the new model?

- Read two papers coming from different communities or on different topics that seem related and try to relate them.
- Explore the connections between machine learning and another field, such as information theory, cryptography or neuroscience.

- **Format.** You should submit your final project using the attached template. In every case, the end result should be a 5 — 8 page report, not including references, plus unlimited space for references, clearly describing what you did, what results you got and what the results mean.

- **Dataset.** If your dateset is not shareable, don't use it because I cannot reproduce your results. You must submit your dataset if it is not publicly available. One of the best places for obtaining "real" data is the repository at University of California, Irvine (`http://archive.ics.uci.edu/ml/index.php`). You can also use synthetic data of your own creation, in which case there is no problem generating a large test set.

- **Code.** The code needs to be submitted along with your report.

- **Results.** You should clearly explain:

  1. What is the problem?
  2. What are existing techniques? Literature review
  3. What is your methodology and improvements?
  4. Numerical experiments for justification
  5. Conclusion

In every case, be sure to explain the meaning of your results. Don't just give a table of results or a dry mathematical formula. Explain what the results mean, and what conclusions can be drawn from them. What did you expect to find? What did you find instead? What are the implications? If you found something surprising, can you think of how it might be explained?

As always, feel free to contact me or Haoyun anytime with questions or difficulties you encounter.